

LEGAL-TECHNICAL INVESTIGATION REPORT

Operation Phantom Swipe

Investigating a Cross-Border ATM & Credit Card Fraud Ring

Case Reference: CC/2026/0147

Assignment 1 — Unit 1: Foundations of Digital Forensics

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NOTE: This report analyses a fully simulated scenario prepared for academic purposes. All persons, devices, card numbers, and account details are fictitious.

1. Executive Summary

This report documents the early-phase digital forensic investigation into “Operation Phantom Swipe,” a simulated cross-border ATM skimming and online credit card fraud ring. The investigation combined physical device seizure (an ATM overlay skimmer), mobile device forensics (a suspect smartphone containing fraudulent banking applications), and analysis of an encrypted archive containing cloned-card data and cash-out network notes.

The scheme involved installation of a card skimmer at a bank ATM in Faridabad, Haryana, capture of card track data and PINs, cloning of physical cards, and monetisation through a network of mule accounts coordinated with overseas (UAE-based) contacts. Digital evidence — call logs, SMS records, GPS traces, and a recovered encrypted archive — corroborates each stage of the fraud lifecycle, from device installation through to cross-border cash-out.

This report sets out the applicable legal framework, the evidence handling methodology followed, the results of forensic search and cryptographic recovery, the cross-border cooperation challenges the case raises, and recommendations for improving law-enforcement standard operating procedures (SOPs) in similar cases.

2. Summary of Crimes and Digital Footprint

2.1 Modus Operandi

The fraud ring operated in four linked stages:

- Physical skimming: An overlay skimmer with a concealed pinhole camera (Exhibit A, device ID SKM-7734-XR) was mounted on an ATM at a bank branch in Sector 21, Faridabad, to capture card track data and PINs.
- Data exfiltration and cloning: Captured data was transferred and used to produce cloned cards; a master list of cloned card numbers was recovered from an encrypted archive on the suspect's storage.
- Fraudulent transaction execution: A fraudulent banking application (“FakeBankLogin”) and a card-data generation tool (“CardGenPro”) recovered from the suspect's phone (Exhibit B) were used to conduct online, card-not-present transactions, with OTPs intercepted via social engineering (evidenced in SMS logs).
- Cash-out and laundering: Proceeds were routed through domestic mule accounts, coordinated via SMS with local handlers, and through recurring calls to two UAE-registered numbers — indicating an organised, cross-border cash-out network.

2.2 Digital Footprint Summary

Evidence Type	Key Finding	Source
Call logs	Repeated contact with two UAE numbers (+971-50-1234567, +971-56-9876543) alongside domestic “handler” numbers	Exhibit B – call_logs.csv
SMS logs	OTP interception and forwarding; explicit references to “batch,” “clone,” and skimmer swap logistics	Exhibit B – sms_logs.csv
GPS logs	Suspect device present at the ATM location during the skimmer installation window (21:40–22:40, 09 Aug 2026)	Exhibit B – gps_logs.csv
Encrypted archive	Cloned card master list and cash-out/mule account notes, recovered via lawful dictionary attack	Exhibit C – secure_vault.zip

3. Cybercrime Taxonomy and Legal Mapping

The scheme spans physical device tampering, data theft, card-not-present fraud, social engineering, and organised money laundering. A condensed mapping to applicable law is given below (full taxonomy and justification in docs/taxonomy.md).

Crime	IT Act, 2000	IPC, 1860	Budapest Convention
ATM Skimming	§43, §66	§420, §468	Art. 2, 7
Card Cloning	§66C	§467, §471	Art. 7
Online / CNP Fraud	§66D	§420	Art. 8
OTP Social Engineering	§66C, §66D	§420	Art. 8
Mule Network / Laundering	§72	§411, §120B	Art. 25 (MLA)
Conspiracy	—	§120B, §34	Art. 25

India is not a signatory to the Budapest Convention; it is referenced here as the standard framework illustrating the type of mutual-legal-assistance mechanism (Art. 25, 29–30) that Indian agencies rely on via bilateral arrangements when a case, as here, has a cross-border nexus.

4. Evidence Acquisition and Handling Justification

4.1 Acquisition Methodology

Both exhibits were seized under standard digital-evidence handling procedure: photographed in place, bagged in tamper-evident packaging, and imaged using a certified hardware write-blocker so the original media was never mounted read-write. SHA-256 hashes were computed immediately upon acquisition and re-verified after analysis; matching hashes (see hashes/sha256_hashes.txt) confirm the exhibits were not altered during examination — satisfying the integrity requirement for admissibility under Indian evidence law (Indian Evidence Act, §65B, certificate of electronic evidence).

4.2 Chain of Custody

A full chain-of-custody form (chain_of_custody/chain_of_custody_form.md) documents every transfer of the exhibits from field seizure through forensic imaging to secure storage, with timestamps and responsible officers recorded at each step. This continuous, documented custody chain is essential to rebut any defence challenge to evidence authenticity.

4.3 Why This Matters Legally

Under §65B of the Indian Evidence Act, electronic records are only admissible with a proper certificate establishing the integrity of the device and process that produced them. The combination of write-blocked imaging, cryptographic hashing at multiple checkpoints, and a documented custody chain directly satisfies this requirement and pre-empts common defence arguments around evidence tampering.

5. Electronic Media Search and Analysis

A forensic string-search pass (scripts/string_search.py) was run across all evidence files, flagging phone-number, GPS-coordinate, card-number-pattern, and keyword matches. The search returned 30 pattern matches across all 8 evidence files, summarised in artefacts/string_search_results.txt. Seven artefacts of direct evidentiary value were extracted and catalogued (artefacts/extracted_artefacts.md), including the overseas contact pattern, GPS placement at the crime scene during the skimmer-installation window, OTP-interception SMS content, and the mule-account references recovered from the encrypted archive.

This staged approach — automated triage search, followed by targeted manual review of flagged files — mirrors standard practice for handling moderate volumes of mixed-format evidence (structured logs, unstructured device dumps, and an encrypted container) without requiring a full manual review of every byte of every exhibit.

6. Cryptography Component

The encrypted archive (secure_vault.zip) was subjected to a lawful dictionary attack (cryptography/dictionary_attack.py) using a 15-entry custom wordlist seeded with common weak-password patterns and case-relevant terms. The password 'skimmer123' was recovered on the seventh attempt in under 5 milliseconds — demonstrating that even data central to an organised fraud scheme was protected with a weak, case-guessable password, a pattern frequently observed in real casework.

The attack was performed only on a forensic copy of the exhibit, never the original; SHA-256 verification before and after the attack confirmed the exhibit's integrity was preserved (see cryptography/cracking_simulation.md and cryptography/attack_log.txt for the full run log).

6.1 Ethical Note: Brute-Force vs Lawful Decryption Requests

Dictionary/brute-force attacks are a proportionate first step against weak, guessable passwords and can be performed entirely on a forensic copy without compelling the suspect to act. Where passwords are strong, the appropriate escalation is a formal decryption direction under IT Act §69, though this raises separate considerations around the right against self-incrimination (Constitution of India, Art. 20(3)) and compliance timelines when the suspect or key custodian is overseas.

7. Challenges in Cross-Border Investigation and Cooperation

- Jurisdictional gap: Key contact numbers and likely cash-out coordination trace to UAE-registered numbers; India is not a Budapest Convention signatory, so formal cooperation depends on bilateral Mutual Legal Assistance Treaty (MLAT) channels, which are typically slower than the Convention's Art. 29–30 expedited-preservation mechanism.
- Data preservation windows: Telecom and payment-processor logs held by foreign providers may be purged before an MLAT request is processed, risking loss of corroborating evidence.
- Attribution across payment rails: Cross-border cash-out via forex conversion and multiple intermediary accounts complicates tracing the ultimate beneficiary of fraud proceeds.
- Divergent legal standards: Evidentiary and procedural requirements (e.g., certification of electronic evidence) differ between India and the cooperating jurisdiction, requiring careful alignment of the evidence package for foreign use.
- Time-zone and language barriers: Coordination with foreign law-enforcement counterparts and telecom providers introduces operational delay at each handoff.

8. Recommendations for Law Enforcement SOP Improvements

- Adopt a standing expedited-preservation request template modelled on Budapest Convention Art. 29–30, usable bilaterally even absent India's accession to the Convention, to reduce foreign-evidence loss during MLAT processing.
- Mandate SHA-256 (or stronger) hashing at three fixed checkpoints — seizure, post-imaging, and pre-analysis — as a uniform SOP across cyber cells, with hash logs attached to the case file by default rather than on request.
- Maintain a centralised skimmer-device signature database (device IDs, firmware fingerprints) to enable faster series-linkage across cases, as was possible here via the recovered device ID.
- Build case-specific wordlists as a standard first step in the cryptography workflow before escalating to §69 decryption directions, given how often case context predicts weak passwords.
- Formalise a joint task-force liaison channel with major cash-out corridor jurisdictions (e.g., UAE) to shorten MLAT turnaround for time-sensitive fraud-ring cases.
- Require dual sign-off (investigating officer + forensic examiner) on every custody transfer, logged in a structured chain-of-custody form from first contact with evidence, not retrospectively.

9. Conclusion

The Operation Phantom Swipe simulation demonstrates a complete early-phase digital forensics workflow: classification of a mixed physical-cyber crime under Indian and international law, forensically sound evidence acquisition with documented chain of custody, targeted media analysis and artefact extraction, lawful password recovery, and identification of cross-border cooperation gaps. The recommendations above are intended to close the specific procedural gaps this case exposed, particularly around cross-border evidence preservation and standardised integrity-hashing practice.

Appendix A: Hash Manifest (Excerpt)

File	SHA-256 (first 24 chars shown)
evidence/password_protected/secure_vault.zip	984b2536a96e639f6931f116a...
evidence/skimmer_device/skimmer_memory_dump.img	edfb1f5dc64492d0529fc74b3...
evidence/suspect_phone/call_logs.csv	b231b4b15beadd22ae15d795d...
evidence/suspect_phone/gps_logs.csv	a2c4fe8053c289fdcaf940f56...
evidence/suspect_phone/sms_logs.csv	fa9fcdcd68f72a674e613507...

Full manifest with complete hash values: *hashes/sha256_hashes.txt*